Subnets vs VLANs

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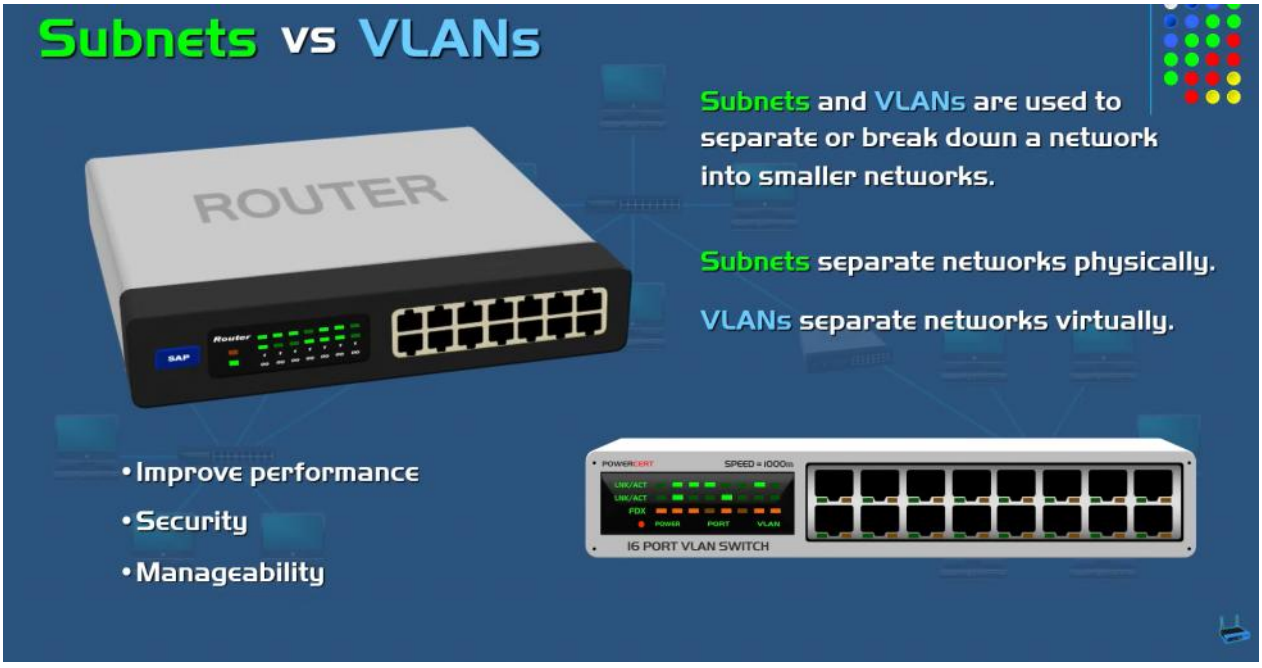
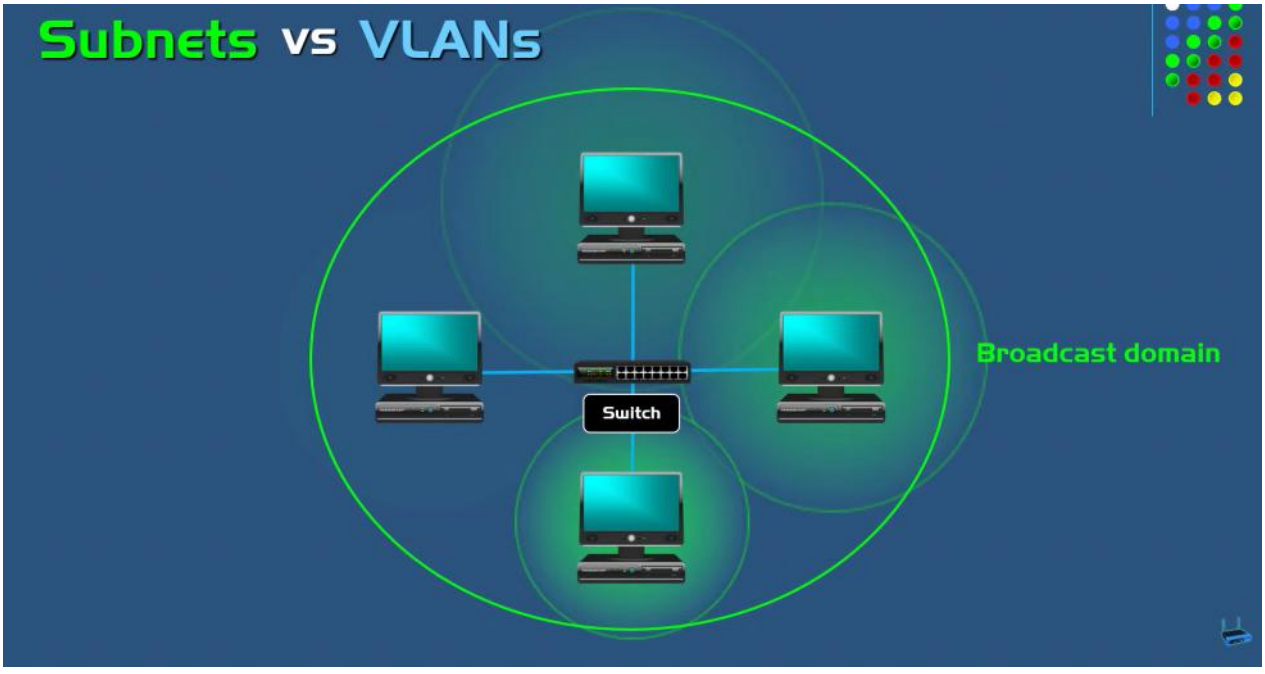
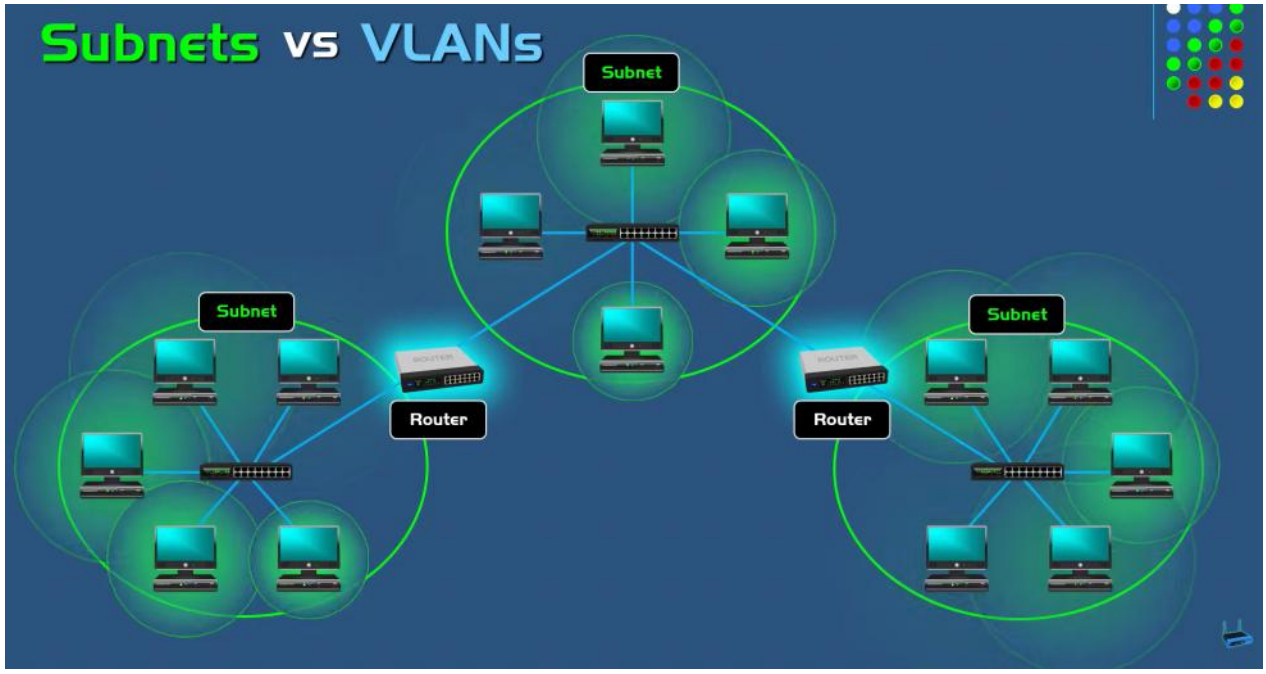
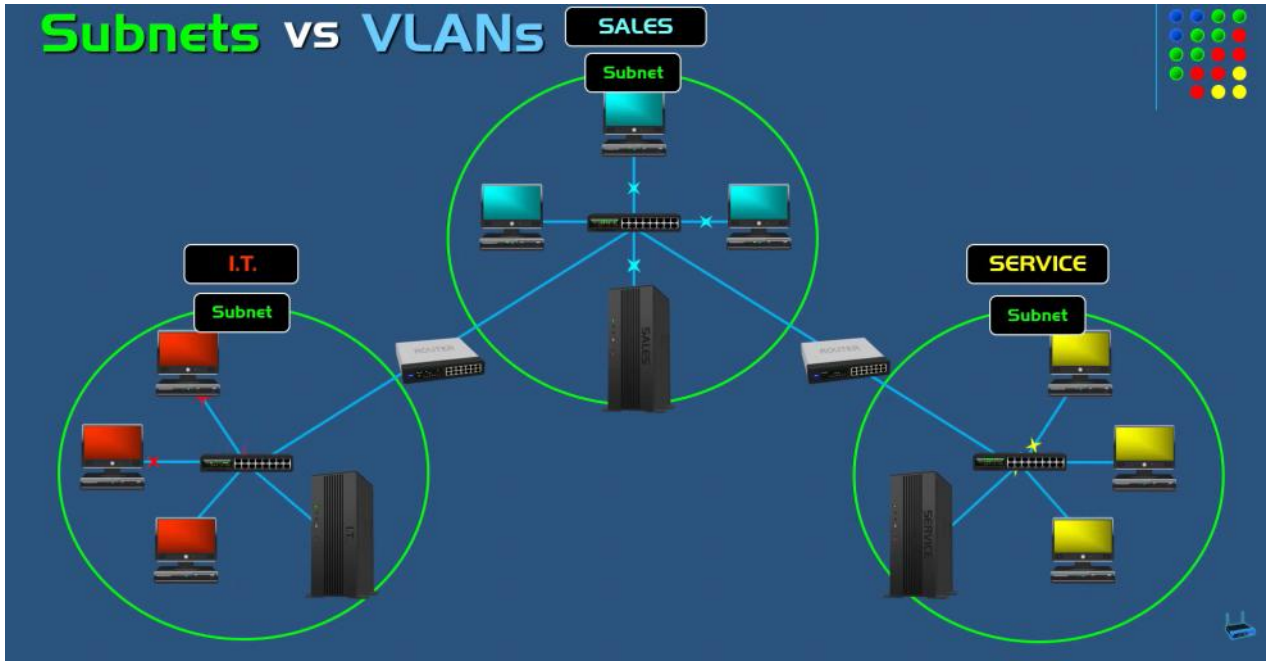
Subnets vs VLANs

Both are used to split a large network into smaller networks for better performance, security, and management.



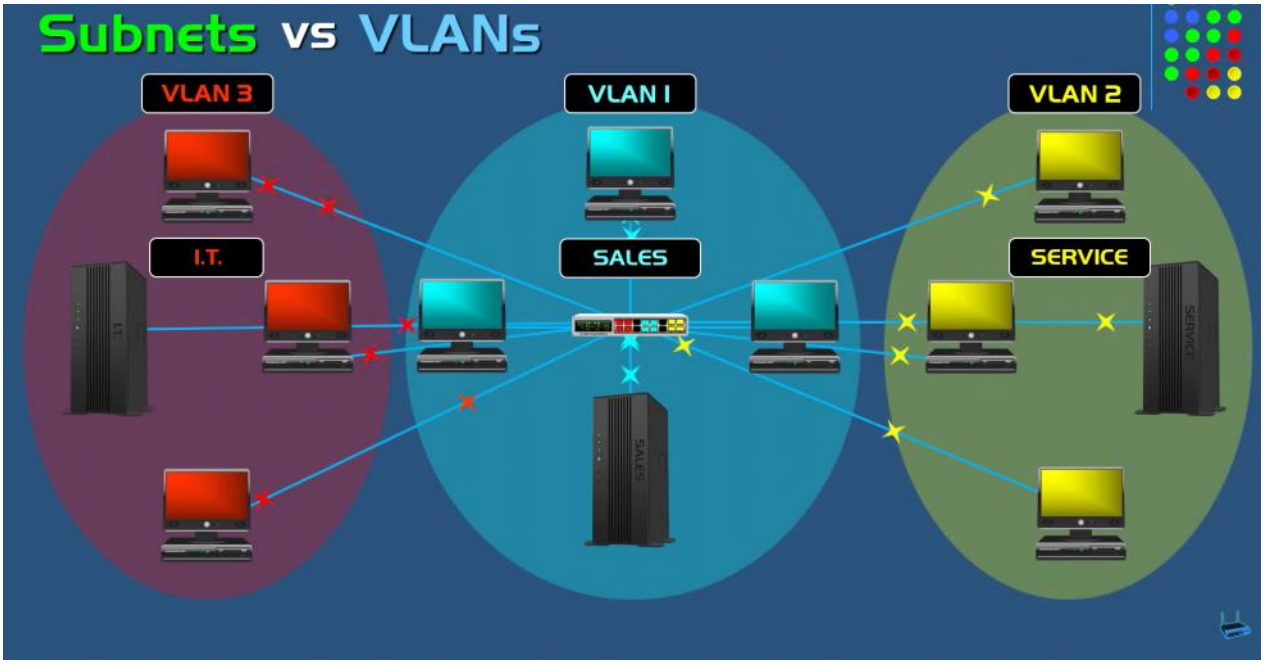






Broadcast stays within the network so It will be easy to manage all the devices as each of them have their own broadcast doma ins.

- **Subnet**
  - Means sub network
  - Created using routers
  - Separates networks physically
  - Each subnet has its own broadcast domain
  - Broadcast traffic does not cross routers



Each of them have their own broadcast domains.

- **VLAN (Virtual LAN)**
  - Created on switches
  - Separates networks virtually without changing cables or adding routers
  - Different ports on the same switch can belong to different networks
  - Each VLAN has its own broadcast domain

- **Key Difference**
  - Subnet uses routers physical separation
  - VLAN uses switches virtual separation

- **Use Case**
  - Both help isolate departments like sales IT service so they cannot access each other's network directly